

Techniques for data cleansing

Readings for today

- Müller, H., & Freytag, J. C. (2003). Problems, Methods, and Challenges in Data Cleansing. Berlin: HUB-IB-164.

Topics

1. Data cleansing
2. Types of anomalies
3. Data quality criterion

Data cleansing

Real data is messy

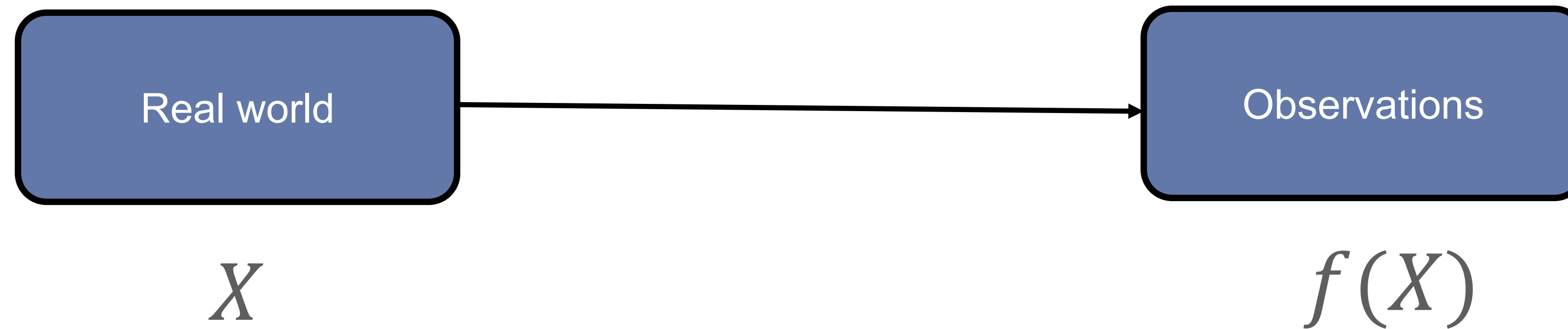
Once you have a tidy data table, there are lots of ways that errors can be present.

- Some are obvious.
- Some are hidden

date	daily deaths	hospitalized
5/7/20	2752	51425
4/29/20	2685	55987
4/15/20	2546	59924
5/5/20	2494	53176
4/21/20	2481	59773
12/1/20	2473	98691
4/14/20	2353	59600
11/25/20	2289	89959
4/16/20	2197	59498
4/30/20	2153	54921
4/17/20	2118	58886
4/22/20	2082	59212
4/11/20	2079	55557
4/28/20	2077	56034
4/10/20	2072	53167
11/24/20	2066	88080

What is data cleansing?

Data cleansing: The identification & accounting for anomalies in your data.



It is assumed that your observations have a veridical mapping to entities in the real world.

Definition of terms

Anomaly: Property of data that renders it an incorrect representation of the world.

Data: Symbolic representation of information.

Tuple: List of discrete values from a finite set.

Feature vector: Collection of observations.

Types of anomalies

Types of anomalies

Syntactical: Errors in labels or formats.

Semantic: Errors in the fundamental value of observations themselves.

Coverage: Gaps in the collection process.

Types of anomalies

Syntactical:

- Lexical errors: Discrepancy between structure of data & format.

Subject ID	Trial	Accuracy	RT
S001	1	correct	incorrect
S001	2	correct	599
S001	3	incorrect	240
S002	1	incorrect	692
S002	2	correct	476
S002	3	correct	301

Types of anomalies

Syntactical:

- Lexical errors: Discrepancy between structure of data & format.
- Domain format errors: Value for attribute does not match domain.

Name	Trial	Cond	RT
Smith, John	1	A	380
Doe, Jane	2	B	599
Smith, Karen	3	A	240
Pair Max	1	A	692

Missing comma in name format

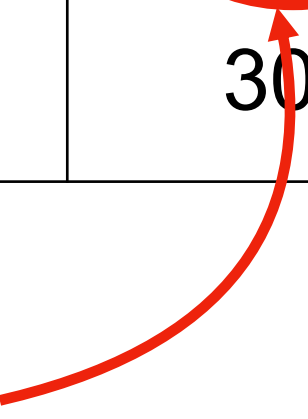
Types of anomalies

Syntactical:

- **Lexical errors:** Discrepancy between structure of data & format.
- **Domain format errors:** Value for attribute does not match domain.
- **Irregularities:** Non-uniform use of values, units, or observations.

Subject ID	Trial	Cond	RT
S001	1	A	380
S001	2	B	599
S001	3	A	240
S002	1	A	692
S002	2	B	0.476
S002	3	A	301

Units change from milliseconds to seconds



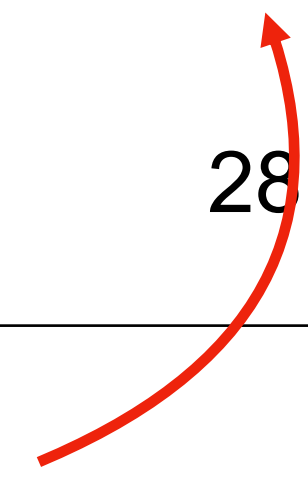
Types of anomalies

Semantic:

- Integrity constraint violations: Value does not match constraints of attribute.

Subject ID	Age	DOB
S001	41	4/14/1979
S002	24	12/25/1996
S003	-1	1/1/1984
S004	28	1/24/1991

Age cannot be negative



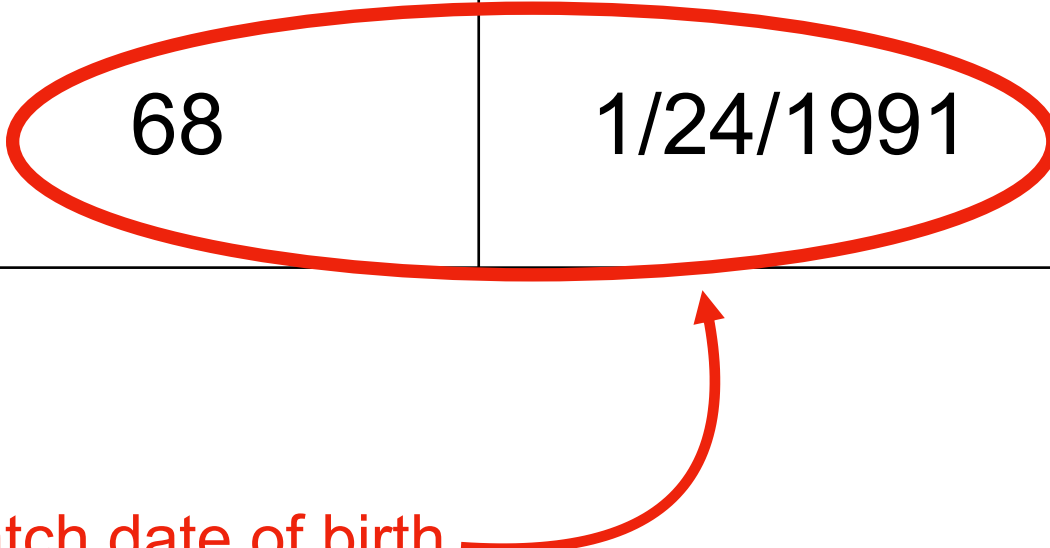
Types of anomalies

Semantic:

- Integrity constraint violations: Value does not match constraints of attribute.
- Contradictions: Values violate a dependency.

Subject ID	Age	DOB
S001	41	4/14/1979
S002	24	12/25/1996
S003	37	1/1/1984
S004	68	1/24/1991

Age does not match date of birth

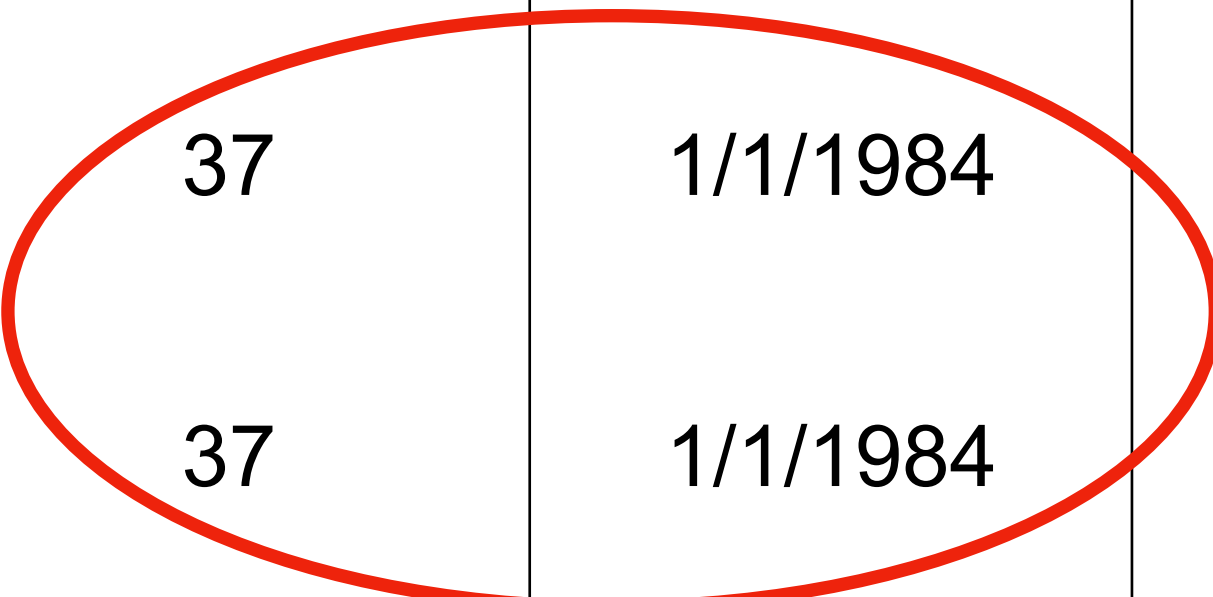


Types of anomalies

Semantic:

- Integrity constraint violations: Value does not match constraints of attribute.
- Contradictions: Values violate a dependency.
- Duplicates: 2 or more data points represent the same thing.

Subject ID	Age	DOB
S001	41	4/14/1979
S002	24	12/25/1996
S003	37	1/1/1984
S004	37	1/1/1984



Likely 2 of the same entry

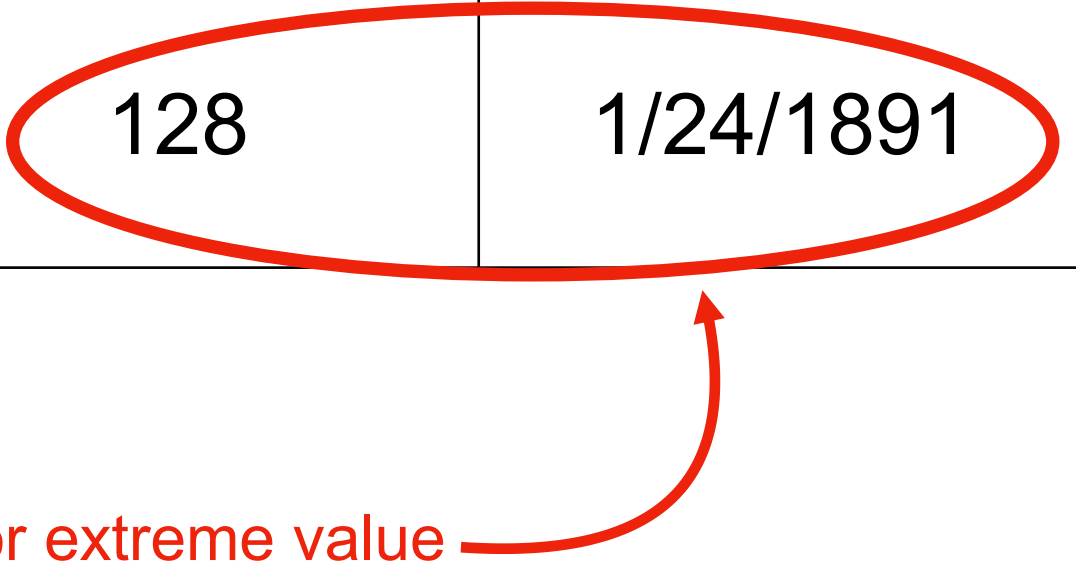
Types of anomalies

Semantic:

- Integrity constraint violations: Value does not match constraints of attribute.
- Contradictions: Values violate a dependency.
- Duplicates: 2 or more data points represent the same thing.
- Invalid tuples: General case for all other semantic errors.

Subject ID	Age	DOB
S001	41	4/14/1979
S002	24	12/25/1996
S003	37	1/1/1984
S004	128	1/24/1891

Outlier or extreme value



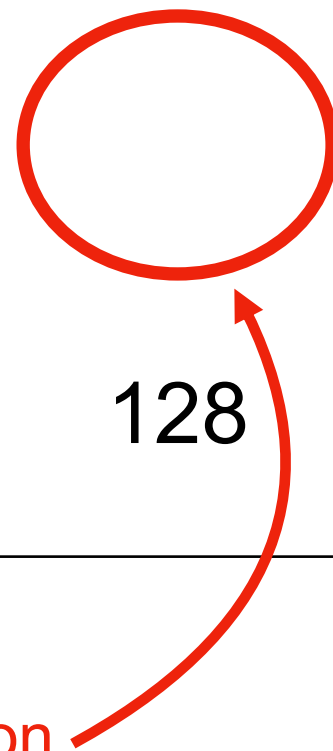
Types of anomalies

Coverage:

- Missing value: Omission of an observation or data point

Subject ID	Age	DOB
S001	41	4/14/1979
S002	24	12/25/1996
S003		1/1/1984
S004	128	1/24/1891

Missing observation



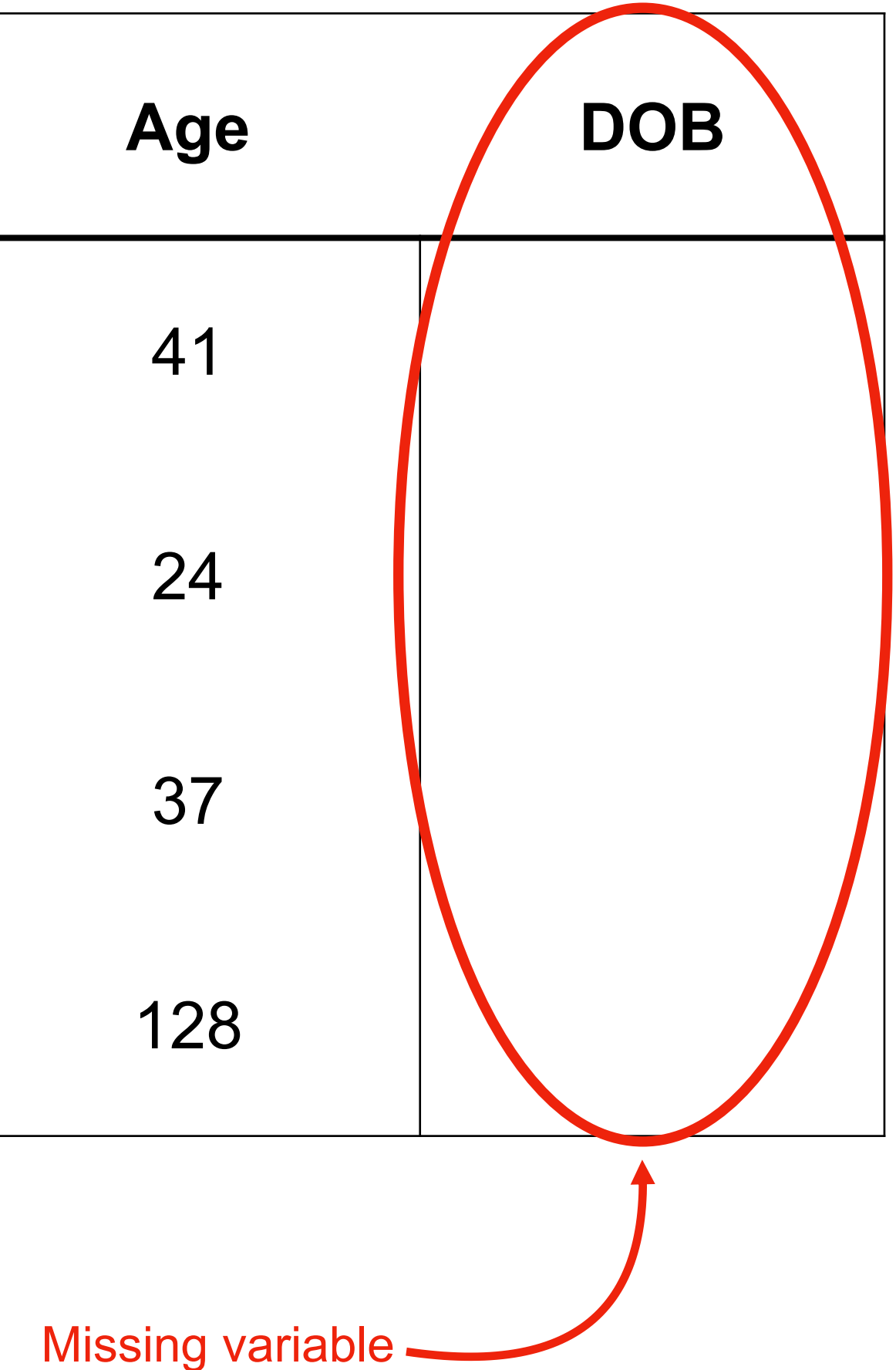
Types of anomalies

Coverage:

- Missing value: Omission of an observation or data point
- Missing tuples: Omission of a full variable or feature.

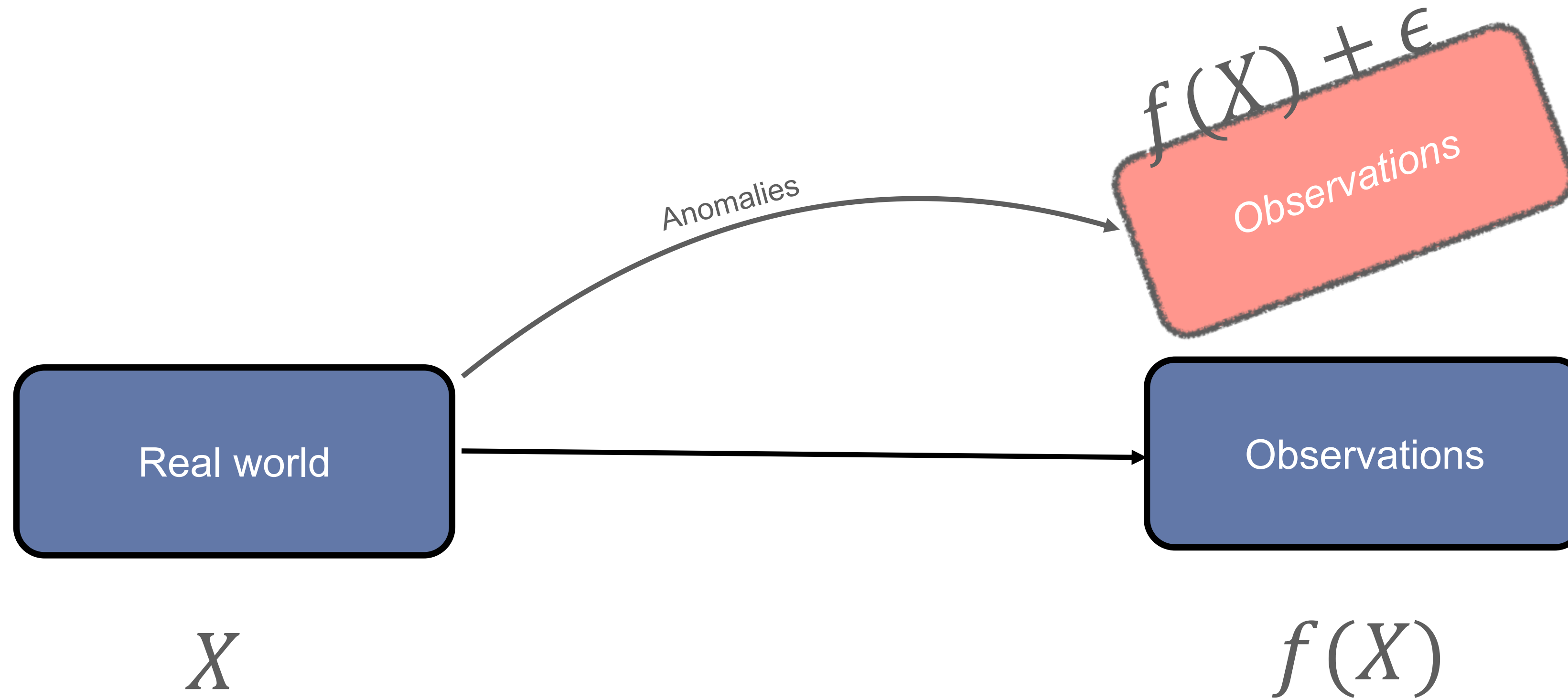
Subject ID	Age	DOB
S001	41	
S002	24	
S003	37	
S004	128	

Missing variable



Data quality criterion

Remember the goal



How do anomalies distort the mapping between world and observations?

Data quality criterion

I. Accuracy: Exact, uniform, & complete representations of the world.

- Integrity: Data set contains representations of all desired aspects of the world. (Semantic)
 - Completeness: All the unique variables & observations are present.
 - Validity: No contradictions or invalid tuples.

Data quality criterion

I. Accuracy: Exact, uniform, & complete representations of the world.

- Integrity: Data set contains representations of all desired aspects of the world. (Semantic)
- Consistency: Data set is uniform & free of contradictions. (Syntactic)
 - Schema conformity: No lexical or domain formatting errors.
 - Uniformity: All observations of the same variable have the same format.

Data quality criterion

I. Accuracy: Exact, uniform, & complete representations of the world.

- Integrity: Data set contains representations of all desired aspects of the world. (Semantic)
- Consistency: Data set is uniform & free of contradictions. (Syntactic)
- Density: For all n observations & p variables, you have exactly $n \times p$ values. (Coverage)

Data quality criterion

I. Accuracy: Exact, uniform, & complete representations of the world.

- Integrity: Data set contains representations of all desired aspects of the world. (Semantic)
- Consistency: Data set is uniform & free of contradictions. (Syntactic)
- Density: For all n observations & p variables, you have exactly $n \times p$ values. (Coverage)

II. Uniqueness: There are no duplicate entries or variables.

Anomaly to quality criterion

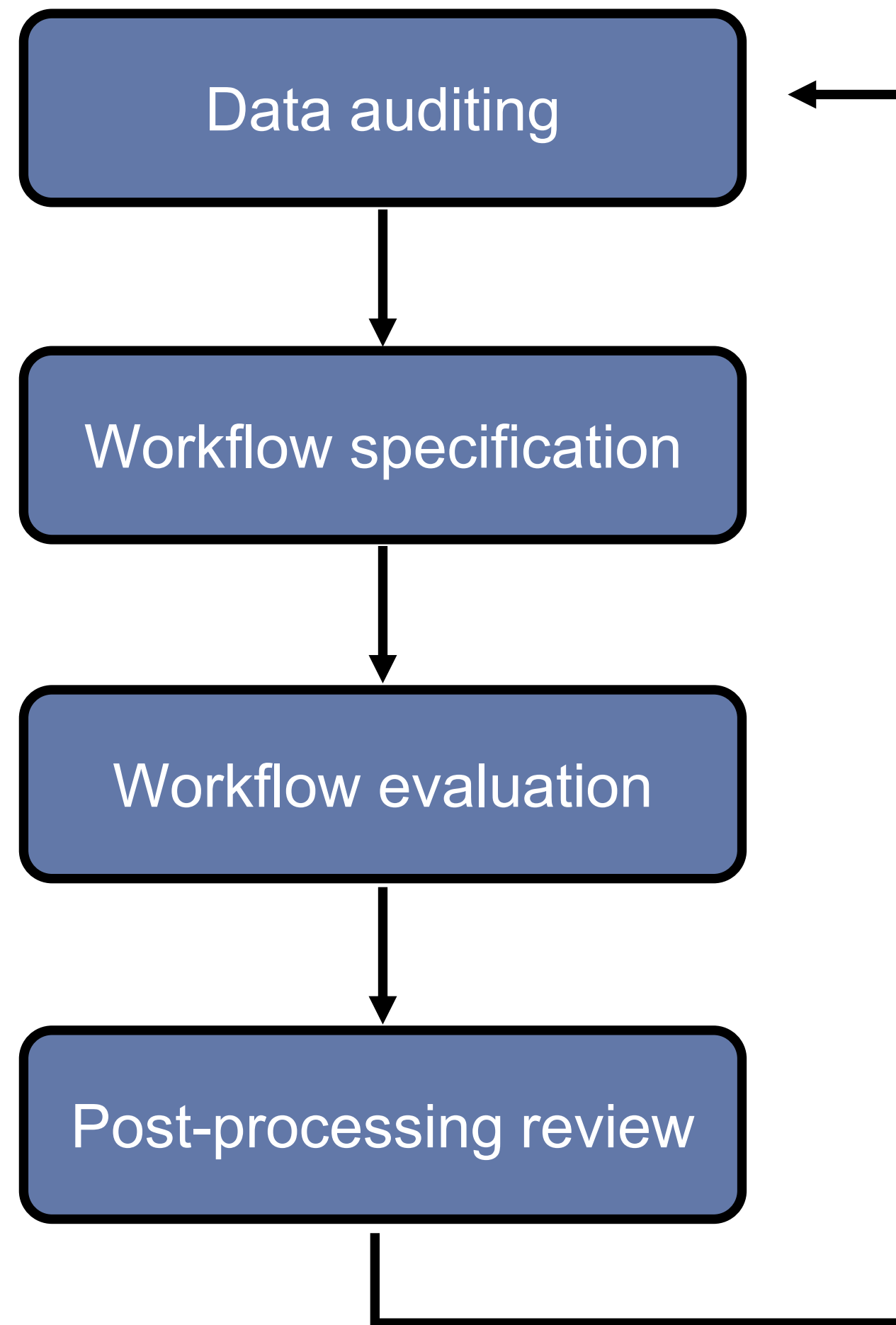
Criterion

Anomaly

	Completeness (all unique vars & obs)	Validity (no contradictions)	Schema Conformity (no lexical/domain errors)	Uniformity (vars have same format)	Density (have n-x-p values)	Uniqueness (no duplicates)
Lexical Errors <i>(RT is 'incorrect')</i>						
Domain format errors <i>(Pain Max not Pain, Max)</i>						
Irregularities <i>(RT is 0.476s not 476ms)</i>						
Constraint Violations <i>(Age is -1)</i>						
Missing Values <i>(Age: {34, _, 27})</i>						
Missing Tuples <i>(Age: {_, _, _})</i>						
Duplicates <i>[Repeated obs]</i>						
Invalid Tuples <i>(x_i: DoB=1891, Age=128)</i>						

Directly impacts
Indirectly impacts

Steps of data cleansing



- Defines a logic for cleansing pipelines.
- Leave original data untouched.
- Automate as many steps as possible

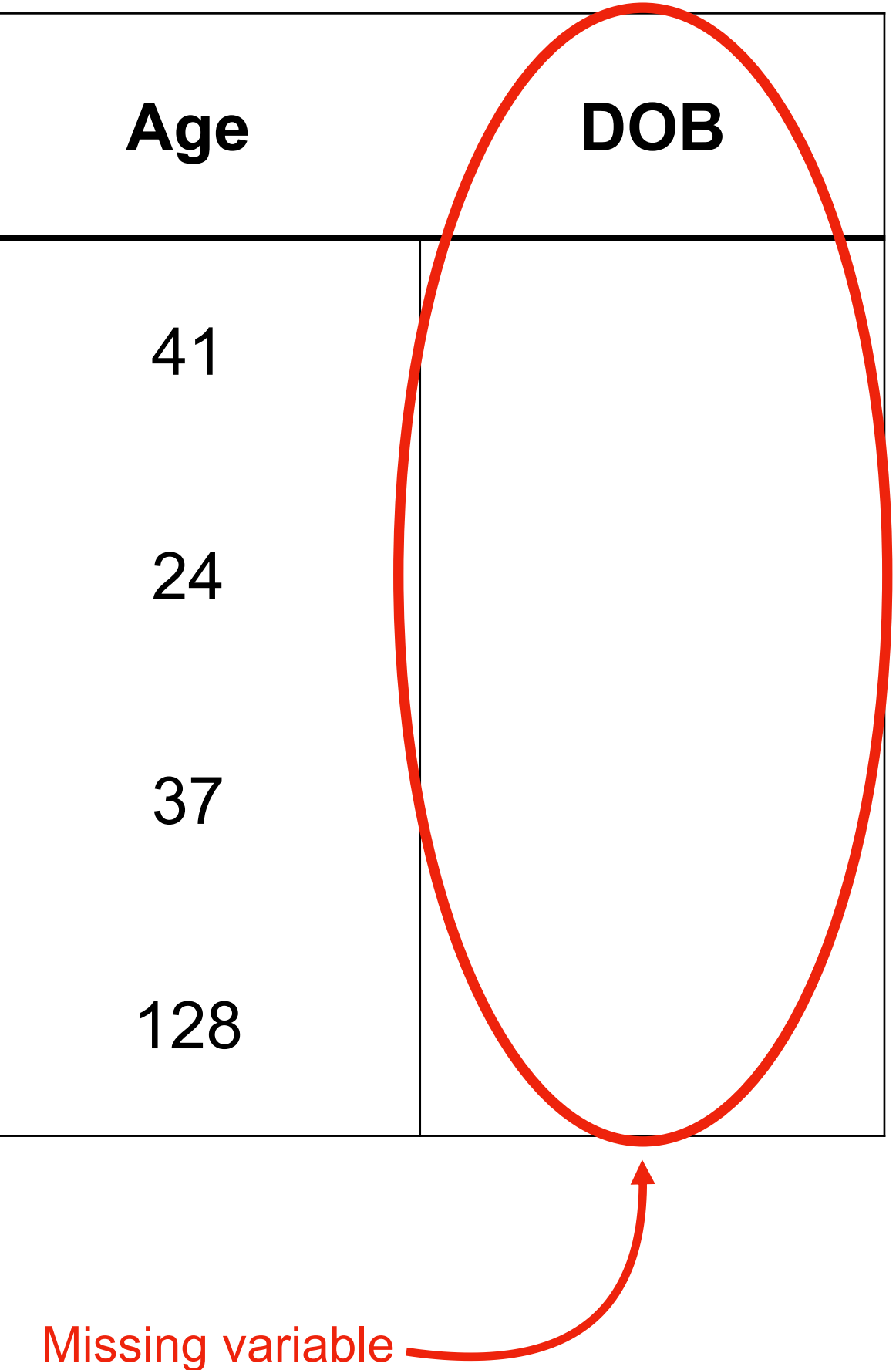
Types of anomalies

Coverage:

- Missing value: Omission of an observation or data point
- Missing tuples: Omission of a full variable or feature.

Subject ID	Age	DOB
S001	41	
S002	24	
S003	37	
S004	128	

Missing variable



Take home message

Developing a formal process for identifying anomalies, correcting identified anomalies, and evaluating for quality makes your data a more veridical representation of the real world.